

Using (4.18) and (4.20), respectively, the *likelihood of the D-vine copula* is given by

$$\begin{aligned} \ell(\theta, \mathbf{u}) = & \prod_{k=1}^n \prod_{j=1}^{d-1} \prod_{i=1}^{d-j} \\ & \times c_{i,i+j;i+1:i+j-1}(C_{i|i+1:i+j-1}(u_{k,i}|u_{k,i+1:i+j-1}), C_{i+j|i+1:i+j-1}(u_{k,i+j}| \\ & u_{k,i+1:i+j-1})), \end{aligned} \quad (7.3)$$

while the *likelihood of a C-vine copula* can be written as

$$\begin{aligned} \ell(\theta, \mathbf{u}) = & \prod_{k=1}^n \prod_{j=1}^{d-1} \prod_{i=1}^{d-j} \\ & \times c_{j,i+j;1:j-1}(C_{i|1:j-1}(u_{k,i}|u_{k,1:j-1}), C_{i+j|1:j-1}(u_{k,i+j}|u_{k,1:j-1})). \end{aligned} \quad (7.4)$$

In particular for D-vine copulas, the conditional distribution function $C_{i|i+1:i+j-1}$ depends first on $\theta_{i,i+j-1;i+j-2}$, while $C_{i+j|i+1:i+j-1}$ on $\theta_{i+2,i+j-1;i+j-1}$. This can be used recursively to determine the complete parameter dependence.

7.2 Sequential and Maximum Likelihood Estimation in Simplified Regular Vine Copulas

The standard parameter estimation method in statistics is maximum likelihood. This involves maximizing the likelihood. Often, it is easier to maximize the log-likelihood instead of the likelihood. Here it is particularly easy, since the likelihood of a regular vine copula is a product of terms, which transforms to a sum, when we optimize the log-likelihood. For hierarchical Archimedean copulas (see for example Savu and Tiede 2010 and Okhrin et al. 2013), this is not possible.

In the case of simplified regular vine copulas in d dimensions with pair copulas with a single parameter, the number of parameters to be maximized is $d(d-1)/2$, which grows quadratically. Therefore it is important to find good starting values for the optimization. Such starting values are given by sequential parameter estimates. We first illustrate this approach in the three dimensional case.

Ex 7.3 (Sequential estimation in three dimensions) The associated likelihood for the three-dimensional case is given in (7.2). In the first step, we find parameter estimates of θ_{12} and θ_{23} by using the bivariate sub samples $(u_{k,1}, u_{k,2})$ and